Mock Assessment - 2 (ArrayList) Without SEB

Java solutions (answer key) | all sample and hidden test cases verified

Q1. Search and Remove the Next Element from an ArrayList (50 marks)

Use **indexOf(name)** (case-sensitive). If it returns -1, print *Not Found*. Otherwise print *Found* and, only if the index is not the last one, remove the element at **index + 1**. In every case print the list with **System.out.println(list)**, which gives the required *[a, b, c]* format.

Java Code (Main.java)

```
import java.util.ArrayList;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String name = sc.nextLine();

        ArrayList<String> list = new ArrayList<>();
        list.add("Aman");
        list.add("Rahul");
        list.add("Sneha");
        list.add("Priya");
        list.add("Karan");

        int index = list.indexOf(name); // case-sensitive
        if (index != -1) {
            System.out.println("Found");
            // remove the element right after the found name (if it is not the last)
            if (index < list.size() - 1) {
                list.remove(index + 1);
            }
        } else {
            System.out.println("Not Found");
        }
        System.out.println(list);
    }
}
```

Test cases checked

Input	Expected output
Rohit	Not Found [Aman, Rahul, Sneha, Priya, Karan]
Rahul	Found [Aman, Rahul, Priya, Karan]
Karan	Found [Aman, Rahul, Sneha, Priya, Karan]
Rah	Not Found [Aman, Rahul, Sneha, Priya, Karan]
sneha	Not Found [Aman, Rahul, Sneha, Priya, Karan]
Sneha	Found [Aman, Rahul, Sneha, Karan]
Aman	Found [Aman, Sneha, Priya, Karan]

Time complexity: O(n), n = 5.

Q2. Even and Odd Count in an ArrayList (50 marks)

Read N, then N integers into an ArrayList. For each value, **x % 2 == 0** counts as even (zero is even, and negative evens such as -2 also give 0); everything else is odd (-3 % 2 is -1, not 0). Print in the format used by

the test cases: *Even:<count>* and *Odd:<count>* with no space after the colon.

Java Code (Main.java)

```
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.util.ArrayList;
import java.util.StringTokenizer;

public class Main {
    public static void main(String[] args) throws Exception {
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
        int n = Integer.parseInt(br.readLine().trim());

        ArrayList<Integer> list = new ArrayList<>();
        StringTokenizer st = new StringTokenizer(br.readLine());
        for (int i = 0; i < n; i++) {
            list.add(Integer.parseInt(st.nextToken()));
        }

        int even = 0, odd = 0;
        for (int x : list) {
            if (x % 2 == 0) even++;   // 0 counts as even; works for negatives too
            else odd++;
        }
        System.out.println("Even:" + even);
        System.out.println("Odd:" + odd);
    }
}
```

Test cases checked

Input	Expected output
5 1 2 3 4 6	Even:3 Odd:2
4 0 -2 -3 7	Even:2 Odd:2
3 1 3 5	Even:0 Odd:3
3 2 4 6	Even:3 Odd:0

Time complexity: O(N). Note: the problem text shows "Even: 3" with a space, but the sample and hidden test cases expect "Even:3" without one, so the code follows the test cases.